Mt. Hood Community College

CS162-H1 Computer Science II

Gresham Campus Hybrid on Zoom • 3/30/2026 – 6/13/2026 • Spring 2026

Faculty Instructor & Faculty Tutor Information

- **Faculty Instructor:** Mitch Priestley
- **Instructor Office:** AC2782
- **Email:** mitch.priestley@mhcc.edu
  Mitch checks email Monday through Friday. Be sure to include your course number in the subject line! He will usually respond to email from your mhcc.edu email within 24 hours.

Faculty Tutors: Jason White and Matt Hess
Tutoring Lab: AC1451 Computer Lab
Email: comptutor@mhcc.edu **Phone:** 503-491-7208

- **Faculty Tutor Meeting Hours** (in person AC1451 or via Zoom)**:**
  Jason: **M 9–1:30 & 2–6; TW 10–1 & 1:30–6**
  Matt: **W 1–6; R 9–1 & 1:30–6; F 10–3; S 10–3**
- **Instructor Meeting Hours (aka Office Hours, Q&A):**
  Mitch is available eight hours a week to meet with students, either in groups or one-on-one, either in person (AC2782) **or** by Zoom (password: **mhcc**):
  - Drop-in hours & locations (virtual by Zoom), through Week 10 (five hours):
    - **M & W 3:00 pm to 3:50 pm — by Zoom (All are welcome)**
    - **M & W 8:00 pm to 8:30 pm — by Zoom (CS162 focus)**
    - **M & W 8:30 pm as needed up to 9:25 pm — by Zoom (All are welcome)**
    - **R 3:00 pm to 3:30 pm — by Zoom (CS260 focus; all are welcome)**
  - By appointment hours & locations (flexible, three hours):
    - Make an appointment by email. Send your availability outside of my published teaching & office hours. Reach out in advance, not at the last minute, if possible.
    - Join office hours in person or by Zoom, using the above link or from Blackboard.
- **Office Phone:** 503-491-7198 (I am not in the office. Please **email** me with questions.)

Course Information

- **Course ID: CS 162-H1 — Computer Science II** (2nd of a 3-course programming sequence)
- **Credits: 4** **Next: Advance in the fall to CS260 and CS250, concurrent enrollment**
- **Prerequisites:** CS161 and MTH095; each with grade "C" or higher, or placement above stated levels.
- **Live Lecture Presentation Time: M & W 6:30 – 8:00 pm** on Zoom. (**passcode: mhcc)**
- **Online Augmentation:** Online worksheets, reading, outside videos, and programming.
- This course fulfills the following: Non-Lab Science
- University transfer equivalent (minimum grade required): PSU(C), OSU(C), EOU, and WOU(C) as CS 162; SOU(C) as CS 257; UO(B-) as CS 211 (Source: Statewide Transfer Articulation Agreement, Computer Science Major Transfer Map Crosswalk, pp. 7 & 20); OIT as CST 126 (Source: OIT).
  Save your syllabus!
  I recommend that you save all your college syllabi for life. They may be required in the future to transfer credits or apply to another institution, even for a master's degree program later in life.

Course Catalog Description

This course is an introduction to object-oriented programming and to software engineering. Students will write programs that use structs, classes, inheritance and polymorphism to manage objects. Topics include recursion, information hiding, testing, and the use of debugging tools.

Learning Outcomes

Upon successful completion of this course, the student will be able to do the following:

1. Apply the principles of object-oriented design in the development of a complete software package, given a set of customer specifications.
2. Develop and implement a plan for testing a program for correctness.
3. Implement advanced language concepts and syntax such as polymorphism and inheritance in class definitions.

Lab Outcomes

1. Communicate the interdisciplinary role of science in current societal issues and the relevance of science to everyday life.
2. Discuss and apply scientific concepts, theories, and models.
3. Select, evaluate, and use discipline-specific information and literature to research a scientific topic.
4. Solve problems independently and/or collaboratively.

Course Activities

- Work individually
- Use a variety of problem-solving strategies.
- Develop and analyze algorithms.
- Design, construct, and test computer programs written in modern C++.

Course Materials

Textbook

Please be prepared to read chapters 8 through 15 and chapter 18 of the course textbook, *C++ From Control Structures Through Objects*, 9th Edition (Grapefruit on cover), by Tony Gaddis. Save money by doing an internet search for a pdf version of the book.

Software

Our students generally use VS Code, available for free download. Be sure to download and install the gnu or g++ compiler, then select it after launch, version 17 or later, preferably 20. You can use any IDE of your choice, and you can get by using Onlinegdb for the first week or two while you install VS Code or your choice of IDE. Use one that lets you do the programming, not an AI-enable one that codes for you—it's much too early for that!

Course Requirements

- **You must turn in coursework during Week 1 to avoid being dropped from class!**
- Lectures – cover textbook material and additional topics
- Publisher Videos – cover select textbook material
- YouTube Videos – cover textbook material
- Discussion posts – initial post and two substantial replies each week
- Online worksheets and quizzes – to verify your comprehension of the textbook
- Lab assignments – short programs written in VS Code to demonstrate topics of week
- Reading – textbook (one chapter per week, plus appendices) and online supplements
- Programming assignments (programs) – to run in VS Code
- Exams – Two Midterm Exams and one Final Exam
- Proficiency Demonstration – Meet with the instructor live (on Zoom or in person) and demonstrate your ability to code before the end of Week 7. Required to pass the course!

Attendance Policy

Log into the course daily to be successful. You earn credit for Zoom **class attendance and participation** by submitting the labs and/or extended labs completed or assigned in each class. The twice-weekly lectures are required, and you are expected to view them live. Attendance and participation are expected and required. If you cannot attend a class, you should reach out to the instructor to make alternative arrangements.

If you get too sick to attend on Zoom live (or have an emergency), then reach out to the instructor in advance to inquire whether it might be recorded for you. It also is good to arrange with another student to take "key notes" for you. Consider making such a pact with a student in class. Do not share labs; they must be completed individually.

Another required live event is the **Proficiency Demonstration**. You will need to meet with the instructor, whether in person or over Zoom, and show the instructor that you can program, using the skills taught in this course.

CS161-H1 Faculty Tutors: Find Them in AC1451 Computer Lab

Jason White has completed all seven CS courses in Mt. Hood Community College's computer science program, and worked as a student aide and tutoring assistant for the previous CS instructor. Jason holds an Associate of Science degree in Cybersecurity and Networking from MHCC, as well as an Associate of Science degree in Electrical Engineering from ITT. Jason is continuing his studies in Information Technology at the Oregon Institute of Technology (OIT), where he is pursuing a Bachelor of Science degree in Cybersecurity. Jason is the founder and president of the Cybersecurity Club at OIT. Jason is a part-time faculty member at MHCC, holding a position as senior Programming Tutor. Jason is also a Non-Commissioned Officer (NCO) in the United States Marine Corps.	 **Jason White, A.S. in Cybersecurity & Networking** (Phi Theta Kappa Honor Society) **A.S. in Electrical Engineering** comptutor@mhcc.edu **M 9–1:30 & 2–6; TW 10–1 & 1:30–6** (Sometimes online for eTutoring) 503-491-7208
	Matt Hess also tutors in C++ programming. **W 1–6; R 9–1 & 1:30–6; F 10–3; S 10–3**

Due Dates, in General

- Lab assignments — 11:00 pm Mondays and Wednesdays
- Discussion Posts – initial post by Tuesday, two substantial replies by Thursday
- Worksheets — as assigned; see Blackboard for each specific due date
- Quizzes & other assignments — 11:00 p.m. on Fridays
- Weekly Programming Assignments — 11:00 pm Saturdays

Late Work Guidelines

All work should be submitted on time and may be submitted early.
This term, work (not exams) is accepted up to 24 hours after the due date without explanation or penalty. Use this sparingly, in case an actual issue arises the next day! Try to post your discussions on time, because other students rely on them to write their responses.

Late work is generally not accepted after 24 hours. Late work undermines the instructor's ability to grade consistently.

Three one-time exceptions beyond the 24 hours are permitted across the entire term for each student, if needed. No explanation as to reason is required. See below.

The student must type the note "Using one-time late assignment waiver" in the Blackboard submission box, specifying which one. The student must keep careful track of what has been used, as the instructor cannot distinguish an accident from an attempt to cheat. Use the submission box on the course home page to keep track. (This means that to use a waiver, you have to do two things, above.)

- **Waiver #1.** One late assignment (up to 24 hours late) may be submitted during the term without penalty.
- **Waiver #2.** One late assignment (up to 48 hours late) may be submitted during the term with 10% penalty.
- **Waiver #3.** One late assignment (up to one week late) may be submitted during the term with 20% penalty.

For other extenuating circumstances be in touch with the instructor, in advance of the deadline when possible, or as soon as is practical, to ask for consideration of an extension or to arrange accommodations. Other than emergencies, extensions are unlikely to be granted, because it is important to keep up with the course as it continues to progress, building on prior skills and learning. Staying caught up is vitally important.

Discussion Posts

This page is important, because discussion posts account for a disproportionately large amount of your grade relative to the time and effort that they take.

Discussion posts are typically scored out of 20 points each week. An initial post and two replies are required. Please avoid providing more than two replies.

The initial post is scored out of 10 points. It should do the following:

- Speak to the prompt (with course content, not encyclopedia or ChatGPT/AI info)
- Provide correct answers
- Explain reasoning
- Cite source (lecture, lab, programming assignment, or textbook)

Note: If you realize that you answered the prompt incorrectly, but learned from others' posts, **you may** use one of your replies to explain your newfound understanding and recover some points that might otherwise be lost to an incorrect initial post.

The two replies are each scored out of 5 points. Each reply should be **substantive** in the following ways:

- Speak to the prompt
- Add new information not contained in your other posts or the post to which you are replying
- Move the discussion forward
- Cite source (lecture, lab, programming assignment, or textbook)

Merely commending the other student with, "Good job! Great post!" and, "I agree, I said this," and repeating will not earn points. **Use your three posts to show what you have learned this week in <u>this course</u> from our textbook, assigned readings, lectures, or labs about the discussion topic. Do not use or paste AI responses.** Cite your sources, at least informally.

Posts/replies that do not contain information from our lectures, labs, and readings will be scored 0, because they do not demonstrate learning the course material and are indistinguishable from AI-generated responses.

Posts/replies that contain outside information should be minimized and <u>must</u> indicate their source.

Posts/replies that do not cite their source will be scored very low.

Replies that are quick, brief, and superficial and do not provide new information will be scored 0 or very low. Your task is not to evaluate other students' posts, though you can offer corrections.

Across your initial post and your two replies, you have three opportunities to show me that you are learning the material from the course. Take all three opportunities to do demonstrate this.

Grading

Each assignment will earn a specified number of points. Those points are added together within each respective category listed in the table below. Then, the percentage is applied to the points. (This is a "weighted" grading system.)

Activity	Percentage
Labs and Extended Labs	20%
Weekly VS Code C++ Programming Assignments	20%
Discussion Posts (Posts are important! And replies must be substantive.)	20%
Worksheets, Quizzes, and other assignments	20%
Exams (3)	20%

Grading Scale: 90%+ A; 80%+ B; 70%+ C; 60%+ D; <60% F

Grades Defined

- A – Excellent
- B – Above Average
- C – Acceptable
- D – Below Average
- F – Insufficient

Weekly Programs - Grading Rubric

- Eligibility for points depends on using the style of our labs and the coursebook (not AI style)
- 10 points for following the style guidance provided (naming conventions, comments, etc.)
- 10 points for meeting assignment specifications (concepts, coding constructs, input/output, program content) – Apply this week's and prior weeks' lessons!
- 5 points for accuracy and completeness of program run

Statement of Originality

All work that you submit for credit or a grade must be your own work, and the statement below must apply to your submission. In addition, the statement below must explicitly appear in a submission during the first half of the course, after you have completed viewing all the assigned videos in the "Avoiding Plagiarism" series.

I understand that all work done at Mt. Hood Community College must be in full compliance with the Student Code of Conduct, including its provision 25 against academic dishonesty. I pledge that this document and all work submitted by me for this course are free of plagiarism. Paraphrases and ideas that are not my own are cited as to their source. I have not attempted to receive academic credit for work generated by ChatGPT, another artificial intelligence, online cheating services, or any other person, entity, or source. My progress toward my college degree represents my own knowledge, skills, and efforts. I have completed viewing the video series on plagiarism assigned by the instructor. I have avoided all forms of cheating. The work I submit is my own, original work.

The consequences for violating this rule (D/F in course) are more serious than merely getting a zero or bad grade on an assignment and can result in dismissal from college and the end of one's college career. Please use restraint and avoid cheating. My least favorite part of the job is the harsh action I take against students who cheat.

When under severe pressure, consider your options:

1. Go to the 1451 Computer Lab tutoring center and have Jason, Adam, Rachelle, or another knowledgeable CS tutor help you with homework, worksheets, or programs.
2. Develop a plan and a schedule to get vital work done.
3. Take a hit on an assignment (but turn something in).

It is always best to avoid cheating.

Extra Credit

You have the option of earning extra credit with your choice of various extra credit assignment opportunities.

You can earn extra credit for each one-page (12-point font, double-spaced) summary of one of the above topics.

- You may submit up to three per week.
- Three could raise your course score by as much as a full percentage point each week.
- Each extra credit can raise your course score by up to 1/3 of one percentage point.

Outstanding Performance

Some extra credit opportunities will be announced via Blackboard or in class. Students who submit extra credit assignments (not just live event credit) and thereby earn above 100.0% in the course overall will receive a Certificate of Outstanding Achievement after conclusion of the course. The custom certificate will be emailed to you as a printable pdf, which you can print to obtain a physical certificate or attach to an email to potential employers. This provides students a chance to garner a tangible endorsement of their superior work ethic.

Communication Policies

- Communicate in a professional manner. This course is a chance for you to model the professional communication skills you would be expected to use in the workplace.
- Negative, inappropriate, or insensitive language will not be tolerated.
- **ALL** communication should demonstrate your knowledge of the English language. ***Correct grammar and spelling are expected, including in discussion forums and email.*** Please avoid text-ese. Any discussion postings that do not use appropriate, professional language will not receive credit. Any such emails will be returned as unreadable. *Write as if you were writing for your boss. Elevate your writing to a collegiate and professional level.*
- ***YOU MUST INCLUDE THE COURSE NUMBER (CS162) IN ALL EMAILS***. If the course number is not in the subject line, the email may be returned to you.

Important Dates

4/11: Optional **Code Party** on campus! Meet the instructor and other CS students in person! 10am – 1 pm, AC2600 (Building 16, upstairs) Free pizza!

5/8: Optional **CS Info Session** (XCR, noon to 1 pm). Complete the **Proficiency Demo** (1:15 pm)! AC1451 (Building 14, downstairs—the computer tutoring lab!)

5/8: Optional **Circuit-Building Party** on campus! Hands-on electrical wiring opportunity to connect switches, lights, and motors/fans. 2 pm.

5/9: Optional **Code Party** on campus! See folks face to face and complete the **Proficiency Demo**! 10am – 1 pm, AC2600 (Building 16, upstairs) Free pizza!

5/25: Holiday — College closed — No class

6/8–13: Final Exams Week — special class schedule (Special office hours will be announced.)

6/8: Final Exam during normal class time.

Other dates may be important to you. Please be in touch with your instructor in advance for accommodation if you need to miss class due to your observance of religious holidays.

Related Special Days of the 2025–2026 Academic Year

Oct. 29 – MHCC Career Fair (Wednesday), 11 am – 2 pm, Vista Dining Room AC2002

Jan. tbd – Portland State Univ.'s MCECS Prospective CC Student Transfer Day 1–4 pm, RSVP

Jan./Feb. tbd – MECOP presentation on internship for next year (app. deadline: March)

Feb. 23 – MHCC's Oregon Transfer Day (Wednesday), 10 am – 1 pm, Vista Dining Room AC2002

May 6 – **MHCC Career Fair** (Wednesday), 11 am – 2 pm, Vista Dining Room AC2002

June 13 – Commencement (Saturday), 10 am

Events in red are those that I highly recommend that you "don't miss."

Extra credits are available for attending the **CS Information Session** (May 8 on campus) and for submitting a self-developed or refined education plan to the Extra Credit Submissions area (2 XC).

Extra credits are available for attending the **Career Fair** (fall and spring term students only) and writing a one-page (12-point font, double-spaced) summary of how many employers you spoke with, whether any were hiring for CS or offered internships related to computer skills in any way, and what qualifications were expected for a computer- or CS-related position. Additional contacts relevant specifically to you are welcome to be included. (2 XC)

Extra credits are available for attending the **PSU MCECS Prospective CC Student Transfer Day** (winter students)and writing a one-page (12-point font, double-spaced) response. (2 XC)

Schedule of Topics & Assignments

Week	Chap.	Monday	Wednesday	Assignments
1	8	VS Code Review – CS161 Topics	Files, Arrays, Search & Sort Solidifying Coding Skills	Discussion Posts & Labs Worksheet Week 1 Programs (2)
2	9	Pointers	Pointers & Dynamic Arrays	Discussion Posts & Labs Worksheet Week 2 Program
3	10	Characters and C-Strings	C-Strings v. String Class	Discussion Posts & Labs Worksheet Week 3 Program
4	11	Structs	Passing Structs Overloaded Operators	Discussion Posts & Labs Worksheet Week 4 Program
5	12	**Midterm 1** Advanced File Operations	Random Access Files	Discussion Posts & Labs Worksheet Week 5 Program
6	13	Classes	Overloaded Constructors w/Default Values	Discussion Posts & Labs Worksheet Week 6 Program
7	14	Operators Overloading Friend Functions	Arrays & Classes	Discussion Posts & Labs Worksheet Week 7 Program
8	15	**Midterm 2** Separate Compilation Helper Functions Inheritance	Polymorphism	Discussion Posts & Labs Worksheet Week 8 Program
9	15, 18	***HOLIDAY***	Classes & Files Linked List – nodes, basic list	Discussion Posts & Labs Worksheet Week 9 Program
10	18	Linked List – full implementation	Linked List – overloaded operators	Discussion Posts & Labs Worksheet Week 10 Program
11		**FINAL EXAM – Monday**	No class on Wednesday	

Notes:

The midterm and final exams will each consist of a coding part and a short answer part (given in Blackboard), including multiple choice, multiple answer, true/false, and fill-in-the-blank.

Live Events

Live Zoom Classes

Every Monday and Wednesday on Zoom from 6:30 p.m. to 8:00 p.m.

Live In-Person AC1451 Computer Lab Tutoring (on campus or by email or Zoom)

Every day, on campus

- Monday – Thursday 9 a.m. to 6 p.m.
- Friday & Saturday 10 a.m. to 3 p.m.
- MHCC Campus, Building 14
  Computer Lab, Room AC1451

Check Jason's and Matt's hours (see page 1) for when each is on-hand to help you write C++ code.
Student Email help: comptutor@mhcc.edu
Student Zoom help: By request
Student Phone help: (503) 491-7208
More tutoring hours and info: https://www.mhcc.edu/student-resources/tutoring

Live Q & A (I encourage you to attend at least one per week if you can)

See page 1 for days of week and times. Conducted over Zoom with instructor or in person or over Zoom with tutor.

Live In-Person Code & Circuit Parties

Code Party: Saturday April 11 from 10 am to 1 pm. Free pizza!
Circuit Party: Friday May 8 at 2 pm in AC1271 Mechatronics Lab!
Code Party: Saturday May 9 from 10 am to 1 pm. Free pizza!

Live In-Person CS Information Session

On campus with Mitch on Friday, May 8, Noon to 1 p.m. AC1451

Live Proficiency Demonstration (Required once per term)

Your choice of either in-person or on Zoom. Show the instructor that you can code. A specific program will be required, for example, a simple counting program using a for-loop. In-person opportunities:

- May 8 at 1:15 pm in AC1451
- May 9 from 10 am to 1 pm in AC2600

To be arranged (by you with the instructor). Take initiative the week of the first mid-term exam to schedule this event to occur before the end of Week 7. Your live proficiency demo is a course requirement. You must complete this task to pass the course.

Notice of Content Change

The instructor may adjust the syllabus, assignments, and class schedule over the term. Students will be notified of changes and revisions through course announcements and/or email. Please check both frequently.

Student Responsibilities

It is assumed that...

- Students will approach their work and their behavior in class in a professional manner.
- Students will do their own work.
- Students will support each other by discussing assignments but will not give answers or share code.
- Students will conduct themselves in accordance with the MHCC Student Code of Conduct.

Participation Requirements

- Weekly participation and submission of assignments is **not** optional.
- If you miss one week of work, you will not be able to make that work up, but may remain in the class.
- If you miss two or more weeks of work, you should consider dropping the class this term & taking it again next term. You simply will fall too far behind to be able to succeed in the class.
- **ALL** assignments are expected be completed at a satisfactory level in order to receive credit for the course.

Technical Support

For issues with Blackboard, contact Online Learning student support services:

- **Call:**(503) 491-7170
- **Email:** onlinelearning@mhcc.edu

Student Support Services

CS-specific tutoring is available in the AC1451 Computer Lab. Programming tutors include Jason and Nicole.

Tutoring and other support services are available in the Learning Success Center (LSC) and AVID Program Center. These offer excellent resources for help with homework, portal and Internet navigation, computer and printer support, study space and College FAQs.

Come to the LSC on the third floor of the library. Drop in Monday-Friday (check website for hours) or call **(503) 491-7108** for an appointment.

Cheating & Plagiarism

Each student is expected to turn in only original work. *<u>Each student is expected to turn in only original work.</u>* Cheating and plagiarism are taken seriously. Any violation of this policy will result in a grade of zero on an assignment that is a copy of or plagiarized from someone else's work. In addition, the violation will be documented in a letter sent to the MHCC Student Conduct Administrator for action. If you turn in work generated by AI, you commit plagiarism. *<u>Each student is expected to turn in only original work.</u>*

Students will receive a zero on an assignment if any of these activities take place:

- the student provides exam questions to other students,
- the student provides homework (written work and programs) to another student,
- the student requests homework answers from another student,
- the student accepts an assignment or program from another student,
- the posts an assignment on the web, social networking site, or discussion forums,
- the student submits work that was obtained from other sources, such as the web, friends, or tutors, or
- the student purchases programs from others.

It is perfectly acceptable to ask for help from others, but the help should be conceptual, not to get the answer to a problem or someone else's program.

Be sure to <u>cite sources</u>.

Course Policies

1. MUTUAL RESPECT.

   Class will be conducted in an atmosphere of mutual respect among students and the instructor.

   a. The instructor pledges to respect students' rights to learn, to express themselves reasonably, and to be treated fairly. He commits himself to providing clear and useful course content, and to constructing fair and relevant examinations.
   b. Students in the course pledge to protect each other's right to learn and to put forth their share of the effort required for the teaching-learning enterprise. Students commit themselves to becoming familiar with course policies and schedule; doing their best to master the material presented; and conducting themselves in an appropriate manner.
   a. Our interactions should recognize and respect potential differences in individuals, including race, language, culture, sexual orientation, gender identity, and other potential differences. Please alert the instructor if greater sensitivity is needed.

2. ACADEMIC INTEGRITY.

   All sources must be referenced in your bibliography with everything you submit. Collaboration should be disclosed at the beginning of your document. You may collaborate with other students and work together on worksheet solutions, but each student must write out his/her/their own work (no photocopies or electronic copying of text, and all writing must be in the individual student's own words. Students have an obligation to integrity in all academic work. In this course, submission of work to be counted toward your course grade automatically implies a personal pledge that you have not given or received unapproved information about the test, copied answers, exchanged unauthorized prior information, or plagiarized written work. A working definition of plagiarism is as follows:

   "*Plagiarism is defined as representing the words or ideas of another as one's own in any academic exercise.*"

   A student found in violation of academic integrity in even the slightest degree will be subject to an academic penalty and/or university disciplinary action. *Be especially careful to avoid electronic copying. It is absolutely a violation to turn in work where you have electronically copied from your e-textbook, from another student, or from any other source. The only exception is that you can electronically copy questions (but* not *answers!). Of course, you may copy/paste your own PC output into your homework answer document, and you may copy from your final paper into your presentation; these are your own work. You can copy your own work! You may not copy someone else's work or provide your work to others.*

   Everyone needs to write their own programs and you are not allowed to share code. You can discuss coding assignments, but you should not show your code to other students or post it to discussions.

3. TEST ADMINISTRATION.

   Early or late examinations or other special arrangements are offered only in cases of documented emergency or conflicts with sanctioned university activities. Students are not to make elective travel or other plans that conflict with the scheduled exams in the courses in which they have registered.

4. CHALLENGING TEST QUESTIONS.
   A student who wants to challenge the validity of a test answer, and who is not satisfied with the instructor's preliminary explanation, may submit in writing the reason(s) a disallowed answer might be considered correct. Challenges should be submitted within 1 week of when exam scores are announced. The instructor will give judicious consideration to the reasoning offered and offer a timely decision.

By enrolling/continuing in this course, you agree to these policies as well as any applicable college policies not mentioned. If for any reason you do not believe that you can adhere to these policies or the expectations listed in this syllabus, then you may drop the course or address your concerns with the instructor immediately.

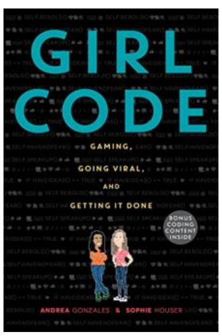
Further Study

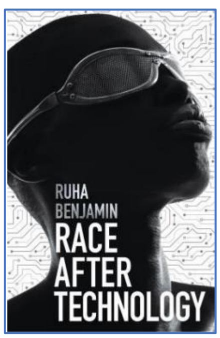
Your next step is to finish the three-course programming sequence: CS161, CS162, CS260. During fall and winter, learn theory (CS250 and CS251) and computer architecture (CS205). You can support your computer education with a study of mathematics, including calculus, linear algebra (MTH261), and statistics (STAT243). Familiarity with Python (ISTM133P, 233P) is helpful. Knowledge of Linux (ISTM140L) is helpful for PSU transfers. Reading and writing skills are also important proficiencies in a CS career. Be able to write requirements, specifications, testing plans, and documentation as well as to create hierarchy charts and flowcharts. Be able to make oral presentations. PSU requires its CS graduates to have MTH261 or 253. Upper-division AI/ML courses may require MTH261 and/or statistics.

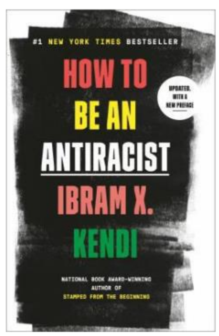
Suggested Reading

Girl Code by Andrea Gonzales and Sophie Houser illustrates how coding can be a form of creative self-expression and how young people can use coding to make a positive impact on the world. An inspirational account written by college-aged coders who gained international acclaim for their program.

Race After Technology ("The New Jim Code") by Ruha Benjamin reveals how passive neutrality in coding and in AI training can inadvertently perpetuate and amplify systemic racism. Discover measures that coders can take to avoid doing so. (See also, *How to Be an Antiracist* by Ibram X. Kendi.)

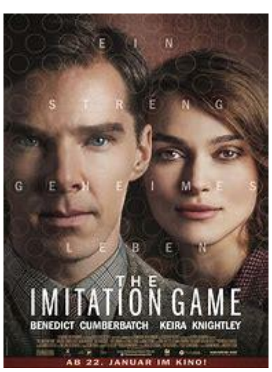






A brief explanation of discriminatory design in technology and AI can be found at https://www.edsurge.com/news/2019-08-20-the-new-jim-code-race-and-discriminatory-design

Suggested Motion Picture



The Imitation Game, directed by Morten Tyldum, is based on the book *Alan Turing: The Enigma* and tells the story of a man renowned for his pioneering work in computer science and in artificial intelligence who helped the Allies win World War II by breaking the sophisticated enemy Enigma code by machine.

Academic Honesty

Cheating, plagiarism, and other acts of academic dishonesty are regarded as serious offenses. Instructors have the responsibility to report any such incident in writing to the Vice-President of Student Development and Services. Depending on the nature of the offense, serious penalties may be imposed, ranging from loss of points to expulsion from the class or college.

Classroom Behavior

Instructors have the responsibility to set and maintain standards of classroom behavior appropriate to the discipline and method of teaching. Students may not engage in any activity which the instructor deems disruptive or counterproductive to the goals of the class. Beepers, pagers, and cellular phones can be a nuisance and are not to be visible or audible in the classroom. Instructors have the right to remove offending students from classes, even Zoom classes.

Americans with Disabilities Act

MHCC is committed to inclusive and accessible learning environments in compliance with federal and state law, including the federal Americans with Disabilities Act (ADA). If you have a disability or think you may have a disability (mental health, attention-related, learning, vision, hearing, physical, or health impacts) please contact the Accessible Education Services (AES) office in AC 2250 or contact (503) 491-6923 or aes@mhcc.edu to have a confidential conversation about academic accommodations. Because accommodations may take time to implement and cannot be applied retroactively, it is important to have this discussion as soon as possible. In addition, individuals with questions regarding ADA accessibility to college public events, please contact (503) 491-6923 or aes@mhcc.edu "

The Accessible Education Services landing page can be found at https://www.mhcc.edu/aes/

Equal Opportunity & Culture of Respect

It is the policy of MHCC to provide an inclusive environment and equal educational opportunity to all persons and prohibits all forms of discrimination based on age, gender, race, color, religion, physical or mental disability, national origin, marital status, sexual orientation, pregnancy, veteran's status, familial relationship, expunged juvenile record, or other status or characteristic protected by law, or association with individuals in such protected status or characteristic. More at https://mhcc.edu/BP-1100/

Affirmative Action

Inquiries regarding application of these and other regulations should be directed to either the College's Affirmative Action Office (503) 491-7200 or TDD, 491-7202, the Office of Civil Rights, Department of Education, Seattle, Washington, or to the office of Federal Contract Compliance Programs, Department of Labor, Seattle, Washington.

Registration Support & Academic Advising

For help with your educational plan, contact the Academic and Advising Transfer Center (http://mhcc.edu/AdvisingCenter/) .

For support in choosing a major or career, contact the Career Planning and Counseling Center (http://mhcc.edu/CareerCenter/) .

For information on student accounts, contact Student Financial Services (http://mhcc.edu/StudentFinancialServices/) .

For advising, counseling or financial aid support, veterans can contact Veteran Services (http://mhcc.edu/VeteranServices/) .

Community Support Services

If you are facing challenges securing food or housing and believe this may affect your performance in the course, please consider the resources linked below for support. If you are comfortable doing so, let me know what you need, and I can direct you to specific college resources.

Get services and support for many different life challenges:

- MHCC Counseling CPCC Personal Counseling (https://www.mhcc.edu/Personal-Counseling)
- Academic Coaching, Navigating College & Career, Scholarships, and Financial Support (http://www.mhcc.edu/successprograms)
- City of Gresham Social Services (https://greshamoregon.gov/social-services/)
- Multnomah Department of County Human Services (https://multco.us/dchs)
- Child Development and Family Support (CDFS) Programs Family Resources (http://mhcc.edu/Family-Resources/)

Read about drug use on campus (https://www.mhcc.edu/DFSCA/)

Basic Needs Hub

1. Need a loaner laptop (two year loans available) with internet?
2. Need internet?
3. Need food?
4. Need rent money to avoid experiencing houselessness?
5. Need utility money to avoid a shutoff?
6. Need car repair money to avoid losing your transportation to college?
7. Need free childcare to attend college?
8. Need help or guidance connecting with or accessing college services?
    - academic advising
    - career planning or job search info
    - emergency support
    - financial aid
    - college food pantry
    - personal counseling and mental health services
    - accessible education and disability services
    - AVID/Learning Success Center
    - Scholars Program (first-generation, first-year MHCC students)
    - work study and student employment at the college
    - tutoring
    - more (all college services)
9. Need help or referrals to community services?
    - Food pantries in the community
    - Help getting an Oregon Trail card (aka SNAP) and Tri-Met discounts (50% off!)
    - Housing referrals

Barney's Pantry

https://www.mhcc.edu/BarneysPantry/

Other Food Resources

SNAP—Supplemental Nutrition Assistance Program: SNAP (also known as Food Stamps) helps you keep healthy meals on the table so you don't have to choose between essentials such as medicine, rent or food. NEW! The application allows you to apply for cash assistance, childcare support and much more at the same time.

Not sure if you qualify for SNAP food benefits? Use this Benefit Estimator.

Links:

Oregon Food Bank: Find Help

More Information on SNAP

Portland Adventist Community Services

Student-Friendly Community Pantries

Gresham/Troutdale Area:

- Snow Cap Community Charities
- Zarephath Ministries
- The Salvation Army Gresham Corps

Portland Area:

- Crossroads Food Bank
- Allen Temple Food Pantry
- Northeast Emergency Food Program
- Esther's Pantry-for HIV+ students
- Harvest Christian Church (only on Wednesdays from 2-4pm)

Sandy/Clackamas area:

- Clackamas Service Center-Food, supplies and more!
- Sandy Community Action Center-Food, supplies and more!

Personal Counseling

https://www.mhcc.edu/Personal-Counseling/

MHCC faculty counselors provide no-cost, short-term, confidential individual counseling for currently enrolled students. Please email counseling@mhcc.edu or call (503) 491-7432 to schedule an appointment.

If you or someone you know is experiencing a life-threatening emergency, please **dial 911** or connect with a crisis hotline for immediate help.

- Multnomah County Crisis Line: (503) 988-4888 or 1-800-716-9769
- Clackamas County Crisis Line: (503) 655-8585
- Clark County Crisis Line: 1-800-626-8137
- Trevor Project: Preventing Suicide Among Gay Youth 1-866-488-7386
- Veterans' Crisis Line 1-800-273-8255, Press 1 or text 838255
- Crisis Text Line: Text "HOME" to 741741 from anywhere in the US

Issues commonly addressed with students by our faculty counselors include:

- **Academic Concerns** related to performance anxiety, perfectionism, underachievement, low motivation, physical or mental disability status
- **Diversity, Equity and Inclusion Concerns** related to impact of oppression, power, privilege, identity, and intersectionality.
- **Developmental Concerns** related to identity development, adjustment to college, and life transitions.
- **Personal Concerns** related to stress and anxiety, depression, anger, loneliness, guilt, low self-esteem, and grief.
- **Relationship Concerns** related to romantic relationship difficulties, sexual concerns, domestic violence, roommate problems and family issues.
- **Other concerns** including effects of trauma, sexual assault, abuse, family history, spirituality, body image, food preoccupation, and healthy lifestyle choices.

Recommended Reading:

A College Mental Health Checklist for Students

Academic Support Programs

https://www.mhcc.edu/Academic-Support-Programs/

Remote Tutoring & Learning Support

AVID/Learning Success Center (LSC)

The AVID/Learning Success Center (LSC) provides technical and academic support to students, including general help and academic coaching, online tutoring, computer skills and technology support.

To learn more, visit mhcc.edu/AVID or contact AVID/LSC staff at AVID@mhcc.edu or 503-491-7331

Mt. Hood Library

The MHCC Library provides online services to all current MHCC students, including access to thousands of ebooks, ejournals, streaming videos, and friendly library staff to help with your research questions.

To learn more, visit mhcc.edu/library or contact a library staff at reference@mhcc.edu or 503-491-7161.

Support Programs that Require an Application

The following programs require an application. See below for details and eligibility criteria.

First-time, First-generation College Students

The Mt. Hood Scholars Program is for students who are:

- The first in their family to attend college
- First-time college students who have completed fewer than 20 credits at MHCC
- Willing to take math and writing in their first year at MHCC

Once enrolled in the program, Mt. Hood Scholars can access the following benefits:

- A designated Student Success Specialist to help them navigate academic advising, financial aid, tutoring, and any other barriers to completing college
- Earn tuition waivers* to help pay for classes by reaching program milestones
- Become part of a community of other students who support each other's success

*Tuition waivers are available for students whether or not they are eligible for federal financial aid.

To learn more and apply, contact Mt. Hood Scholars staff at successprogram@mhcc.edu or 503-491-7471.

SNAP Eligible Students

The **SNAP Employment and Training Program (STEP)** is for students who are:

- Receiving or eligible to receive SNAP benefits for themselves (formerly called Food Stamps)
- Not receiving TANF welfare benefits

Once enrolled in the program, STEP participants can access the following benefits:

- Gap funding for college costs
- Work closely with a Student Success Specialist to create a financial and education plan
- Regular communication with the Student Success Specialist to help ensure educational success
- Help navigating college and community resources, and referrals to resources

To learn more and apply, contact STEP program staff at successprogram@mhcc.edu or 503-491-7471.

Students Planning to Transfer to a Four-year College or University

TRIO Student Support Services (TRIO-SSS) is for students who:

- Plan to graduate from MHCC and transfer to a four-year university
- Meet **at least one** of these criteria: low-income, documented disability, or first person in their family to attend college
- Meet the residency requirements to receive federal student aid

Once enrolled in the program, TRIO-SSS participants can access the following benefits:

- Academic support, career planning and transfer advising
- Guidance on affording college including help navigating financial aid and applying for scholarships
- Tutoring
- Organized visits to four-year colleges and universities

To learn more and apply, visit mhcc.edu/TRIO-SSS/ or contact TRIO program staff at TRIO.Programs@mhcc.edu or 503-491-7688.

Women Starting or Returning to College

Transitions and its sister program, Transiciones (offered in Spanish) are for women who are new to college, have been out of school for a while, or who need extra support to help them succeed in college.

Once enrolled in the program, Transitions or Transiciones participants can access the following benefits:

- Become part of a supportive community of MHCC staff, mentors, and peers
- Comprehensive career planning and college preparation assistance
- Help navigating college services and resources
- Help developing skills including time management, study strategies, and applying for scholarships

To learn more and apply, visit mhcc.edu/Transitions or contact Transitions/Transiciones program staff at 503-491-7680.

Land Acknowledgement

We honor the Indigenous people on whose traditional and ancestral homelands we now stand: the Multnomah, Kathlamet, Clackamas, Tumwater, Watlala bands of the Chinook, the Tualatin Kalapuya, and many other Indigenous nations of the Columbia River. It is important to acknowledge the ancestors of this place and to recognize that we are here because of the sacrifices forced upon them. In remembering these communities, we honor their legacy, their lives, and their descendants.

Artificial Intelligence Addendum

Write your own work. Anything you submit for credit must be written by you! If you submit work created by ChatGPT, then you're not earning college credit. That's plagiarism, which is a serious violation of college and university rules.

The remainder of this Addendum is how ChatGPT says it.

Ethical Use of AI Tools in Academic Work[i]

In this course, we value the pursuit of knowledge, critical thinking, and intellectual growth. It is our belief that education is a journey, not just a destination. Therefore, we encourage you to actively engage in the learning process, which includes taking ownership of your academic work.

The Dangers of Shortcuts

In recent years, advancements in artificial intelligence have introduced tools that can assist with various tasks, including essay writing, coding, and problem-solving. While these tools can be incredibly powerful and efficient, their use in academic work raises important ethical considerations.

Academic Integrity Matters

Using AI tools to complete assignments, essays, or coding projects, and submitting them as your own work, not only undermines the educational experience but also erodes the core values of academic integrity. It's akin to taking a shortcut through a beautiful forest, missing the opportunity to explore, learn, and grow along the way.

A Degree's True Worth

Obtaining a college degree is a significant achievement that reflects your dedication, hard work, and acquired knowledge. By relying on AI tools to complete assignments, you risk devaluing the very credentials you seek to earn. It's like receiving a law degree merely for delivering documents, without having the legal expertise that should accompany it.

Learning for Life and Career

Education is not just about passing exams; it's about preparing for life's challenges, acquiring problem-solving skills, and fostering a deep understanding of your chosen field. The use of AI tools for academic work may offer short-term convenience but can hinder your long-term growth as a thinker and problem solver.

The Choice is Yours

I encourage you to resist the temptation of quick fixes and embrace the learning process fully. The journey of exploration, experimentation, and genuine engagement with your studies is where true growth and achievement lie.

Conclusion

This course is designed to nurture your intellectual curiosity and encourage you to develop your own unique insights and skills. Please remember that the value of your education extends far beyond grades and credentials; it is a lifelong journey of self-discovery and personal growth.
We expect all students to uphold the highest standards of academic integrity. Any form of plagiarism or unethical use of AI tools will be addressed in accordance with the college's academic honesty policy. Thank you for your commitment to your education, and I look forward to a rewarding and intellectually stimulating semester together.

Appendix A: Understanding How to Avoid Plagiarism – Video Series (Required)

Avoiding Plagiarism

Plagiarism is representing the words or ideas of another as your own or failing to cite the source of words or ideas.

Using someone else's ideas in your paper without giving credit can result in a 0 on the assignment, an F for the entire course, and/or lead to institutional consequences such as suspension or expulsion from the college.

Please watch the following videos to avoid plagiarism (8 short videos):

1. https://youtu.be/SjqegPFnlEY?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
2. https://youtu.be/Uk1pq8sb-eo
3. https://youtu.be/uQhVDH9p7aU?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
4. https://youtu.be/oiM0x0ApVL8?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
5. https://youtu.be/DhMl3eIcGbI?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
6. https://youtu.be/opp259YvaoE?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
7. https://youtu.be/HlN8gD22gFs?list=PLjBMY3HggCpCI-AOVnhpMSei1G4kNhGvI
8. https://youtu.be/N4j6ILb_hBo

Artificial Intelligence

Submitting text or code (programs) generated by AI for class credit is plagiarism on the basis that it is submitting work that is not your own.

Appendix B: CS Topics Explained – Video Series

Video tutorials from the publisher and others on YouTube illustrate key concepts from the textbook.

CS162 publisher videos

Week 1 — Chapter 8: Searching and Sorting Arrays

- The Binary Search
- The Selection Sort
- Solving the Account Validation Modification Problem

Week 2 — Chapter 9: Pointers

- Dynamically Allocating an Array
- Solving the Pointer Rewrite Problem

Week 3 — Chapter 10: Characters, C-Strings, and More about the string Class

- Writing a C-String Handling Function
- Using the string Class
- Solving the Backward String Problem

Week 4 — Chapter 11: Structured Data

- Creating a Structure
- Passing a Structure to a Function
- Solving the Weather Statistics Problem

Week 5 — Chapter 12: Advanced File Operations

- Passing File Stream Objects to Functions
- Working with Multiple Files
- File Encryption Filter Problem

Week 6 — Chapter 13: Introduction to Classes

- Writing a Class
- Defining an Instance of a Class
- Employee Class Problem

Week 7 — Chapter 14: More about Classes

- Operator Overloading
- Class Aggregation
- Solving the NumDays Problem

Week 8 — Chapter 15: Inheritance, Polymorphism, and Virtual Functions

- Redefining a Base Class Function in a Derived Class
- Polymorphism
- Employee and ProductionWorker Classes

Week 10 — Chapter 18: Linked Lists

- Appending a Node to a Linked List
- Inserting a Node in a Linked List
- Deleting a Node from a Linked List
- Solving the Member Insertion by Position Problem

CS162 YouTube videos

Week 1, Review & Chapter 8, Searching and Sorting Arrays

Review Default Arguments | C++ Tutorial (8:21)
Review Use A Ternary Operator As An Lvalue | C++ Tutorial (5:47)
Review `swap()` Standard Library Function | C++ Tutorial (4:30)
Review Namespaces | C++ Tutorial (10:45)
Optional:
Review Why `using namespace std` is Bad in Big Programs| C++ Tutorial (optional, 15:13)

Week 2, Chapter 9, Pointers

C++ POINTERS (2020) - How to create/change dynamic arrays at runtime? (13:46)
C++ POINTERS (2020) - Introduction to C++ pointers (for beginners) (11:10)
C++ POINTERS (2020) - What is a void pointer? (for beginners) (15:12)
C++ POINTERS (2020) - How to use pointers and arrays (for beginners) (12:51)
C++ POINTERS (2020) - What is a dynamic two-dimensional array? (17:48)
SMART POINTERS in C++ (for beginners in 20 minutes) (optional, 24:31)
Function Pointers for beginners! How and when to use Function Pointers? (optional, 22:26)

Week 3, Chapter 10, Characters, C-Strings, and More about the `string` Class

C++ Lesson 13.0 - Null - Terminated Character Arrays (10:14)
C++ Lesson 13.1 - Built-in Functions for Null-Terminated Character Arrays (11:21)
C++ Lesson 13.2 - String and Character Manipulation (17:40)
String Concatenation | C++ Tutorial (7:48)

Week 4, Chapter 11, Structured Data

C++ Lesson 12.0 - The Struct (10:32)
C++ Lesson 18 - Enumerations (6:39)
C++ Lesson 14.2 - Passing Streams to Functions (9:45)

Week 5, Chapter 12, Advanced File Operations

C++ Binary File I/O (20:22)

Week 6, Chapter 13, Introduction to Classes

C++ Lesson 15.0 - Object-Oriented Paradigm (9:09)
C++ Lesson 15.1 - Defining Class Functions (12:52)
C++ Lesson 15.2 - Accessor and Mutator Functions (8:22)
C++ Lesson 15.3 - Const Member Functions (2:13)
C++ Lesson 15.4 - Friend Functions (3:12)
C++ Lesson 15.5 – Constructors (11:28)
Destructor Basics | C++ Tutorial (9:12)
Declare And Initialize An Array Of Objects | C++ Tutorial (9:13)

Optional:
Deleted Functions | C++ Tutorial (optional, 5:55)
Defaulted Functions | C++ Tutorial (optional, 2:59)

Week 7, Chapter 14, More about Classes

C++ Lesson 15.6 - Structs vs Classes (3:02)
C++ Lesson 15.7.0 - Overloading Operators (11:44)
C++ Lesson 15.7.1 - Multiplication and Chaining (5:26)
C++ Lesson 15.7.2 - IsEquals Operator == (4:14)
C++ Lesson 15.7.3 - Negation and Not Operators (optional, 3:17)
C++ Lesson 15.7.4 - Constructors for Automatic Conversion (4:17)
C++ Lesson 15.8.0 - Special operators: Assignment Operator = (4:09)
Optional:
C++ Lesson 15.8.1 - Special Operators: Bracket Operator [] (optional, 2:42)
C++ Lesson 15.8.2 - Function Evaluation Operator () (optional, 6:18)
C++ Lesson 15.9 - Static Members (optional, 6:06)
C++ Lesson 15.10 - Template Classes (optional, 16:01)
How To Implement Method Chaining | C++ Tutorial (optional, 6:19)

Week 8, Chapter 15, Inheritance, Polymorphism, and Virtual Functions

Separate Compilation
C++ Lesson 10.0 - Multiple Files (9:32)
C++ What is class inheritance? [1] (6:43)
Demystifying C++ Access Specifiers: Protected Keyword & Class Access Specifiers [2] (8:25)
C++ Inheritance: constructors and destructors in base and derived classes [3] (8:22)
C++ Inheritance: Redefining base class functions [4] (5:25)

Week 9, Chapter 15 (continued), Inheritance, Polymorphism, and Virtual Functions

C++ Inheritance: Class Hierarchies [5] (7:15)
C++ Polymorphism and Virtual Member Functions [6] (2:12)
C++ Abstract base classes and pure virtual functions [7] (7:08)

Week 10, Chapter 18, Linked Lists

Introduction to Linked Lists - Data Structures and Algorithms (21:59)
C++ Tutorial - LINKED LISTS (6:13)
Define A Copy Constructor To Create A Deep Copy Of An Object | C++ Tutorial (17:05)
Object Assignment | C++ Tutorial (16:22)

Appendix C: Crash Course Computer Science – Video Series (Required)

Platform: YouTube
Channel: CrashCourse
Playlist: Learning
Category: Computer Science

It is important to assimilate the information in these videos as you progress through the CS program. The goal is to complete the series before entering CS205, when you will need to know all this information.

CS160 –

Required: Videos 1 – 10

CS161 –

Required: Videos 11 – 20

Recommended: 1 – 10 if you have not already viewed these!

CS162 –

Required: Videos 21 – 30

Recommended: 1 – 20 if you have not already viewed these!

CS260 –

Required: Videos 31 – 40

Recommended: 1 – 30 if you have not already viewed these!

CS205 –

Required Videos: A list will be provided.

Recommended: 1 – 40 if you have not already viewed these!

Appendix D: Mathematical Fluency and Algorithms

Solutions exist in many forms. Two forms of solutions are solution sets and algorithms.

The following are two well-known examples of exercises in mathematical fluency and problem solving.

Example 1:

Three consecutive integers sum to 21. (1) What are the integers? (2) Demonstrate the solution. (3) Model the problem. (4) Demonstrate an algorithm that leads to a solution.

Example 2:

A person with a fox, a chicken, and a sack of grain must use a boat to cross a pond carrying one item at a time. The fox cannot be left unattended with the chicken, nor the chicken with the grain. (1) What is the solution (in English, step by step)? (2) Demonstrate the solution. (3) Model the problem. (4) Model the solution. (5) Demonstrate an algorithm that leads to a solution. (6) Demonstrate that the algorithm does not violate the constraints of the problem.

Appendix E: Citation Syles (APA Citation Style & IEEE Citation Style)

A citation style is a style for citing sources and formatting your papers. We are concerned that you cite your sources. You may use any popular style, such as APA, IEEE, or MLA. You may also use another computer-industry specific style, such as ACM or CSE. We want you to learn one style and apply it consistently in citing your sources and listing your sources at the end of your paper.

We recommend that you use one style for all your classes. Therefore, if another course requires you to use APA or MLA, then we encourage you to use that same style for this course, because your goal should be to learn one style completely during your first year of college.

IEEE style is built into Word, is a computing industry style, and is similar to ACM and CSE. Therefore, if you are not using another style for a different course, then we recommend IEEE. We recommend that you use a style that is built into MS Word, because it will automatically number your sources and renumber them for you, saving you time and reducing numbering errors. Word can also be used to automatically create your references list at the end of your paper in the correct order and format. The choice of style is up to you.

For APA style, see Introduction, Workshop, in-text citations, and reference list basics.

For IEEE style, see Overview, citations, and reference list. (Complete style guide manual is here.)

Our course will focus on your use of citations and a reference list at the end of your paper. Other elements of the style will not be the focus on this course.

Please submit your papers double-spaced, using a 12-point font and 1” margins. Titles and headings may be bold and may be 14-point.

APA Citation Style

Mt. Hood Community College (MHCC) generally uses **APA (American Psychological Association) citation style**, particularly for science, social science, and health-related fields.

Here's a breakdown of key aspects of APA style:

1. In-Text Citations

- **Narrative Citation:** Incorporate the author's name into the sentence and follow with the year in parentheses (e.g., Jones (2020) argues...).
- **Parenthetical Citation:** Place the author's last name, year of publication, and page number in parentheses at the end of the sentence (e.g., (Smith, 2019, p. 25)).
- **Multiple Authors:** Use "et al." for three or more authors in in-text citations (e.g., (Smith et al., 2020)). The abbreviation *et al.* is Latin and means "and the others," referring to people, versus *etc.* or *et cetera*, which means "and other things" or "and the rest," referring to things.
- **Direct Quotes:** Include the page number when quoting directly (e.g., (Smith, 2019, p. 25)).

2. Reference List

- The reference list is titled "References" and is placed at the end of the paper.
- Entries are arranged alphabetically by the first author's last name.
- Each entry includes the author's last name, first initial, year of publication, title of the work, and publication information (e.g., journal, publisher, DOI).

3. Examples of APA Citations (Reference List Entries)

- **Journal Article:**
    - Lastname, F. M., Author2, B. B., & Author3, C. C. (Year). Title of article. Title of Periodical, volume number (issue number), pages.
    - Example: Flores-Cruz, Z., & Allen, C. (2011). Necessity of OxyR for the hydrogen peroxide stress response and full virulence in Ralstonia solanacearum. Appl Environ Microbiol, 77(18), 6426-6432.
- **Book:**
    - Author, A. A. (Year). Title of book. Publisher.
    - Example: Smith, J. (2018). The History of Science. Academic Press.
- **Website:**
    - Author, A. A. (Year, Month Day). Title of page. Website Name. Retrieved from URL
    - Example: Smith, J. (2019, March 15). The Importance of Citation. My Website. Retrieved from https://www.example.com

This page was produced by experimental generative AI and was then edited moderately by the instructor.

Appendix F: Full Path to a Computer Science Education

MHCC offers seven courses in computer science:

CS160 Computer Science Orientation	offered Fall, Winter, Spring, plus even-year Summer
CS161 Computer Science I	offered Fall, Winter
CS162 Computer Science II	offered Winter, Spring
CS260 Data Structures	offered Fall, Spring
CS250 Discrete Structures I	offered Fall only
CS251 Discrete Structures II	offered Winter only
CS205 Systems Programming & Architecture	offered Spring only

The three-course programming series consists of CS161, CS162, and CS260. Language: C++.
The two-course mathematics of computer science series consists of CS250 and CS251.

CS250 & CS251 cover the mathematical aspects of computer science.

CS205 explores computer architecture (hardware) from a software standpoint and touches on low-level programming in C language and assembly language.

Here is the usual sequence of computer science courses:

Year One			Year Two		
Fall	Winter	Spring	Fall	Winter	Spring
CS160	**CS161**	**CS162**	**CS260***		**CS205**
			CS250	**CS251**	

*CS260 can be delayed until Spring, but CS260 and CS205 are both very hard courses.

Here is a common alternative sequence:

Year One			Year Two		
Fall	Winter	Spring	Fall	Winter	Spring
CS161	**CS162**	**CS260**	**CS250**	**CS251**	**CS205**
	CS160*				

*CS160 can be taken any term.

Students generally acquire a math education at the same time that they are progressing through their computer science courses. Calculus, required for the degree, can be waived as a pre-requisite for some courses.

Associate of Science Transfer in Computer Science degree (AST-CS):

For transfer to PSU, OSU, UO toward a B.S., CS   More here   Career Info
For transfer to EOU, SOU, WOU toward a B.S., CS   Career Info
CS Course Descriptions

Here is a sequence of immersive computer science starting summer quarter:

Summer	Fall	Winter	Spring
CS161	**CS160** **CS250**	**CS162** **CS251**	**CS260** **CS205**

End Notes

[i] The page titled “Ethical Use of AI Tools in Academic Work” was composed by OpenAI’s ChatGPT-3.5 language model on 9/17/2023 in response to instructor Mitch Priestley’s prompt: “I am a college instructor. My classes begin in a week. I need a page in my syllabus generated by ChatGPT (that's you) that explains why using ChatGPT or other, superior AI tools to do their essays, writing, and CS coding is unethical, undermines their education, short-circuits their learning, and devalues educational credentials (college degrees). It's like awarding a law degree to a courier who delivers a lawyer’s documents to a client. You should write a compelling, emotional tale, painting an analogy that would make a student feel like they'd be playing the idiot to cheat using Google or AI to find answers or to compose replies that they would turn in as their own work for credit.” The bot began its reply, “Certainly, I can help you craft a compelling message for your syllabus that explains the ethical concerns associated with using AI tools for academic work. Here's a sample text that you can consider including…” and the rest was included in this document’s AI Addendum after only light editing.